Predicting Coronary Heart Disease from BRFSS Survey Answers

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Code: `ML-Project-1` on GitHub

***Abstract*—We predict heart attack or coronary heart disease (MICHD) from 321 answers to the 2015 BRFSS health survey (8.8% positives). We compare 13 methods and submit an ensemble of two Newton logistic regressions, one with feature products and one followed by boosted trees: F1 = 0.4356 in 5-fold cross-validation, 0.446 on AIcrowd. The threshold and the encoding matter more than the model, and we keep out a column that reveals the label.**

I. Introduction

The BRFSS is a yearly US telephone health survey [1]. From its 2015 answers (328,135 training and 109,379 test people) we predict MICHD: the person was told they had a heart attack, angina or coronary heart disease.

With 8.8% positives, answering "healthy" for everybody is 91% accurate, so accuracy is useless. The challenge uses F1, the harmonic mean of *precision* (the share of flagged people who are positive) and *recall* (the share of positive people we flag). Flagging everybody gives F1 = 0.1623, the best rule on one question (self-rated general health) 0.3088.

We contribute a survey-aware pipeline with an ablation of its steps, 13 methods compared on the same folds, an ensemble that beats the best single model in every fold, and a measured label leak.

II. Models and Methods

A. Data and preparation

Our exploratory analysis found three problems. *Codes are not values:* 7 usually means "don't know", but "retired" for the employment question, so we wrote one rule per column from the codebook. *Missing values are everywhere and informative:* an average column is 47% missing, because of optional modules and of skip logic (a follow-up question is not asked when it does not apply); people who answered the diabetes module have 21.9% MICHD, the others 7.9%. *Many columns repeat others.* Train and test are distributed alike.

Clean: special codes become "missing" or 0, the 14 mixed-unit columns get one unit, and skipped questions are filled with "no". *Select:* drop 41 columns that do not describe the person (ids, dates, survey weights) and 30 duplicates, then keep 124 of the 236 left with four rules. The main one drops 89 columns that remove almost none of the uncertainty about the label (*mutual information* below 1 of 431 millibits). *Fill:* add an "is missing" column per group of columns missing together when missingness shifts the MICHD rate by $\geq$3 points; impute the median (numbers) or the mode (yes/no, categories). *Encode:* yes/no becomes 0/1; categories, ordinal answers whose risk is not monotone (e.g. smoking status) and the binned days of poor physical health become one 0/1 column per value with at least 500 people (*one-hot*); then every column is standardized. This gives 336 inputs.

B. Models

Required methods: linear regression by gradient descent (GD) and by stochastic GD, least squares (pseudo-inverse, as $X^\top X$ is singular), ridge regression, and logistic regression by GD with and without the penalty $\lambda\|w\|^2$.

Other optimizers of the regularized logistic loss: adaptive-step GD (Barzilai–Borwein [2]), UniXGrad [3], Sophia [4], and *Newton's method*, which uses the curvature of the loss to set each step: about 10 steps reach the minimum, with no step size.

Trees on 32 bins per column [5]: a random forest [6] and gradient boosting [7], [8], [9], i.e. small trees each fitted to the errors left by the previous ones.

Logistic boosting fits the logistic regression with Newton ($\lambda = 10^{-4}$), then starts the boosting from its log-odds $x^\top w$ instead of a constant. The linear part captures the additive effects (age, general health) exactly; 400 trees (depth 3, learning rate 0.05, $\geq$100 people per leaf) learn what is left.

Newton with expanded features adds squares of the non-binary columns, $\log(1 + x - \min x)$ of the continuous and count ones, and the products of all pairs of the 20 columns most correlated with the label (190 products, chosen inside each fold): 619 columns, $\lambda = 10^{-6}$. Hand-made clinical features (counts of conditions, products such as age $\times$ sex) were also screened, first with a score test on the fitted model, then in CV; none improved F1 (not tried with the trees, which already combine columns by their splits).

Ensemble and threshold. The two models above have the highest CV F1 (Table I) and capture interactions differently (fixed products vs. trees), so they may err on different people and averaging can cancel part of their errors. The submission averages the two probabilities, $p = a\,p_{\text{Newton}} + (1-a)\,p_{\text{boosting}}$, with the weight a also tuned to maximize F1 ($a = 0.35$), and predicts MICHD when $p \geq t$. With 8.8% positives,

Table I
F1 IN 5-FOLD CV (MEAN ± STD), BASIC PREPARATION (CODES TO MISSING, >50%-MISSING COLUMNS DROPPED, MEDIAN, ONE-HOT) VS. OURS. †WEIGHT AND THRESHOLD SET ON THE OTHER FOLDS; SO EVALUATED, ITS PARTS SCORE 0.4336, 0.4338.

Method	Basic	Ours	Setting (ours)
Linear regr., GD	0.4184	0.4211 ±0.0018	γ 0.1, 300 it.
Linear regr., SGD	0.3996	0.4084 ±0.0028	γ 10^{-4}, $5{\cdot}10^{4}$ it.
Least squares	0.4179	0.4212 ±0.0025	–
Ridge regression	0.4181	0.4216 ±0.0026	λ 10^{-3}
Logistic regr., GD	**0.4270**	0.4311 ±0.0017	γ 0.5, 500 it.
Reg. logistic, GD	**0.4270**	0.4308 ±0.0013	λ 10^{-4}, γ 0.3
Reg. logistic, Newton	–	0.4323 ±0.0020	λ 10^{-4}, $\leq$20 it.
Reg. logistic, adaptive GD	–	0.4322 ±0.0019	λ 10^{-4}, 300 it.
Reg. logistic, UniXGrad	–	0.4318 ±0.0016	λ 10^{-4}, 200 it.
Reg. logistic, Sophia	–	0.3928 ±0.0050	γ 0.02, 3000 it.
Random forest	–	0.4158 ±0.0018	100 trees, depth 14
Gradient boosting	–	0.4298 ±0.0024	200 trees, depth 3
Logistic boosting	–	0.4338 ±0.0020	400 trees, depth 3
Newton, expanded features	–	0.4339 ±0.0016	619 columns
Ensemble (submitted)†	–	**0.4356** ±0.0020	$a = 0.35$

$t = 0.5$ flags almost nobody, so t maximizes F1 on training predictions, never on the scored rows ($t = 0.206$).

C. Evaluation

All experiments share 5 stratified folds (seed 1): pipeline, model and t are fitted on four folds, F1 is measured on the fifth. We report mean ± standard deviation (about 0.002, so smaller differences are noise) and in how many folds a variant beats its reference. Hyperparameters were searched in stages on small grids. a (step 0.05) and t are chosen on out-of-fold probabilities, then both models are refitted on all training rows. `run.py` rebuilds the submission in about 11.5 minutes.

III. RESULTS

Baselines (Table I). Our preparation improves each required method by 0.0027 to 0.0088. Logistic beats linear regression by about 0.01: the squared loss also penalizes predictions that are correct by a large margin, so easy rows pull the fit, while the logistic loss models the probability of a 0/1 label and ranks people better for the threshold. Three optimizers of the same loss end within 0.0005, while Sophia, the only stochastic one, stays at 0.3928. Trees alone do worse.

Ablation (Fig. 1). The threshold dwarfs everything: at $t = 0.5$, logistic boosting scores 0.2395 instead of 0.4338. Among data steps only one-hot encoding is large; the selection adds nothing measurable, and each cleaning step not shown changes F1 by less than 0.001. Squares and logarithms change nothing (0.4323 to 0.4324), products do not help logistic boosting (-0.0005), and clinical features lower F1 by 0.0004 to 0.0013.

Submissions. The ensemble weight is stable across folds (0.35 to 0.40). On AIcrowd the ensemble scores 0.446 (logistic boosting alone 0.445). Our best online score, 0.452, used the leaking column discussed below (0.449 as a rule on the ensemble): the test performance of our submission is 0.446.

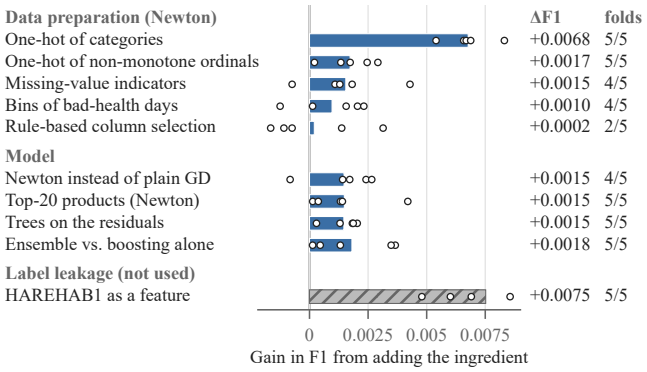


Figure 1. Gain in F1 from adding each ingredient, all else fixed (bars: mean of the 5 folds; dots: each fold; "folds": folds with a gain). Data rows use Newton. Not shown: tuning t instead of 0.5 adds 0.1943.

IV. DISCUSSION

What drives the prediction. The signal agrees with known cardiac risk factors. The answers most informative about MICHD (mutual information) are self-rated general health, age, employment status and blood pressure medication; a rule on general health alone already reaches F1 0.31. Among our clinical products, only age × sex and age × smoking carry signal beyond the model (score test), too little to raise F1. These are associations in self-reported answers, not causes: some (medication, poor health) may follow the disease.

Where the gain comes from. All reasonable models end between 0.43 and 0.435: the limit is probably the information in the answers, not the model. Products and trees give Newton the same +0.0015 and their gains do not combine, so they likely capture the same interactions. We did not check that the ensemble's parts err on different people.

A column that reveals the label. `HAREHAB1` asks about rehabilitation *after a heart attack*, so it is asked only to people who had one: all 773 training rows where it was asked are positive. As a feature it adds 0.0075 in all folds, and as a rule it lifts the ensemble to 0.4414. We do not use it: it reveals the label, and it exists only after the event.

Limitations. The selection rules saw the labels of all training rows, so CV is slightly optimistic (unmeasured; the selection itself adds only 0.0002). The ablation uses Newton (data) and logistic boosting (model), not the ensemble, which was too slow to retune for every variant; since the ensemble adds only 0.0033 to Newton alone, the gains should be similar for it. $\lambda = 10^{-6}$ is at its grid edge, and there is no nested CV.

V. SUMMARY

Careful encoding and an ensemble of two Newton logistic regressions give F1 = 0.4356 in CV and 0.446 on AIcrowd.

Threshold and encoding matter most, and a leaking column would add score, not usefulness.

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